

Jayesh RL

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Education

Dayananda Sagar University, Bengaluru Expected Graduation: May 2027
B.Tech in Computer Science (Artificial Intelligence & Machine Learning) | GPA: 9.45 / 10
Relevant Coursework: Data Structures & Algorithms, Machine Learning, Artificial Intelligence, Statistics & Applications, Object-Oriented Programming, Computer Organization

Technical Skills

Programming: Python, Java, JavaScript, SQL, C
Frameworks & Tools: React, Next.js, Node.js, Spring Boot, TensorFlow, Firestore, FastAPI, Git, Jupyter
ML & AI: Deep Learning, Time-Series Analysis, Foundation Models, LLMs, Attention Mechanisms, Feature Engineering
Signal Processing: EEG / ECG Analysis, Quantization, Spiking Neural Networks
Soft Skills: Research Mindset, Analytical Thinking, Collaboration, Continuous Learning

Experience

Lead Software Developer Jun 2025 – Aug 2025

Humans Care Foundation (HCF), Bengaluru

- Developed a fully responsive and SEO-optimized web platform using React, Next.js, TypeScript, and Firebase.
- Achieved 100+ active users within the first month of deployment.
- Managed production deployment with SSL security, domain configuration, and real-time database integration.

Backend Engineering Intern Jan 2025 – Feb 2025

Why Digit System Private Limited, Bengaluru

- Built and optimized backend services using Java and Spring Boot, reducing code redundancy by ~30%.
- Designed and tested 10+ RESTful APIs with JWT-based authentication and robust error handling.
- Optimized complex SQL queries, reducing query latency by up to 40%.

Projects

Predictive Maintenance — Digital Twin-Based Data Science System

- Built a production-grade time-series data science pipeline for real-time vehicle telemetry, including validation, statistical cleaning, feature engineering, and health state computation.
- Implemented a digital twin rule engine with trend-based prediction to estimate time-to-failure and enable explainable, proactive maintenance decisions.

ThreatPeek — AI-Powered Security & DevSecOps Platform

- Built an AI-augmented security scanner to detect exposed API keys, secrets, and web misconfigurations across web assets.
- Implemented a custom crawling and detection engine for HTML, JavaScript, and HTTP header security analysis.
- Integrated LLM-based contextual threat summaries and severity scoring to provide actionable remediation insights.

Research2Text — RAG-Based Research Intelligence System

- Built a local-first RAG system using semantic chunking, vector embeddings, and citation-aware retrieval for research papers.
- Designed a multi-agent system to extract methodologies and generate validated Python/PyTorch code from academic papers.

Research & Publications (In Progress)

Personalized Cardio Equation for Individualized ECG Modeling – Patent Research

- Designing an AI-driven parametric equation to model individual cardiac behavior from ECG signals.
- Focused on personalized, interpretable cardiac modeling for adaptive cardiovascular diagnostics.

Ethical Hiring Bias Detection Using Machine Learning and Explainable AI – Under IEEE Review

- Developed ML models for automated hiring with fairness constraints across gender and age groups.
- Applied fairness metrics and XAI techniques (SHAP, LIME) to enable transparent and accountable hiring decisions.

Leadership & Activities

DSU × Tempête – Club Coordinator Aug 2024 – Present

- Coordinated high-impact technical and departmental events engaging 100+ students.

Hackathons

- 1st Place – DASHACK Data Visualization Hackathon (DSU)
- 2nd Place – Quizathon, IEEE DSU Student Branch
- TECNOCOGNITION – Certificate of Appreciation (\$100 USD)
- Top 15 – 24-Hour Hackathon, TEDxDSU (Apr 2025)